

Mill Spindle - Troubleshooting Guide

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Symptom	Possible Cause	Corrective Action
Spindle not turning and at Power On: A Warning appears " The spindle may be rotating freely without power... "	The software version is outdated.	Check the software version to be 100.20.000.1200 or higher. Contact your Haas Factory Outlet to have the software updated.

Noise during rigid tapping.	A NGC software error in version 100.17.000.xxx that caused a noticeable growling noise during rigid tapping, particularly with coarse pitch threads.	Update software to the latest version.
	Problems with the spindle encoder.	Look for runout, loose mounting screws or shaft clamps, and any other obvious problems. Spindle Non-Contact Encoder (NCE) - Troubleshooting Guide - NGC
The Spindle is seized or it won't turn.	The spindle has been crashed and now damage.	Review alarm history for indications of a crash, check for visual damage to the spindle
	There is heavy side loads causing the cap to gall the shaft.	Check for clearance all the way around with a piece of 0.001" shimstock; loosen the cap, re-center and tighten.
	The spindle lubrication system is not functioning correctly.	Inspect the spindle lubrication system.
	Spindle installed incorrectly	Check Motor coupler for damage or indentations at the pin interfaces. Contact local HFO about issue including relevant pictures of the spindle to motor coupler.
The spindle temperature exceed 140° F (60° C)	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.
The spindle makes an unusual noise.	The spindle drive belt is worn or damaged.	Check the spindle drive belt for damaged.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.

The surface finish shows chatter, or the spindle vibrates. Alarm 966 EXCESSIVE PART/TOOL IMBALANCE.	The Main Processor PCB standoff hardware is loose or the Control Cabinet mounting hardware is loose or causing a bind.	Tighten the Main Processor PCB standoff hardware. Loosen the Control Cabinet mounting hardware, snug the hardware, then torque down the hardware.
	The tool or tool holder may be too long for the application.	Longer tools are less stable than shorter tools. Always use the shortest tool possible, and remember that a 10% reduction in the length-to-diameter ratio results in a 25% increase in tool stiffness. Unless absolutely necessary, the tool's stick-out from the holder should not be more than 3 times longer than its diameter.
	The spindle drive belt is worn or damaged.	Check the spindle drive belt.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The toolholder is fretting.	Verify the length of the tool with the manufacturer's specifications.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct. Refer to: VMC - Spindle - Drawbar - Force Measurement
	The fixture does not support the part.	Adjust the cutting load or program.
	The tool is unbalanced.	Balance the tool holder with the cutting tool in it. If you cannot balance the tool assembly, replace it with one that you can balance.
	Excessive Endmill Runout - Mill.	Tool runout should not exceed 0.0003" (0.076 mm). To check the tool runout, place a dial indicator on the tool, and then rotate the spindle by hand.
	The spindle bearings are damaged.	Perform a vibration analyzer.
	The spindle is not balanced.	Contact your HFO to balance the spindle.

Tools stick or do not come out of the spindle easily.	The tool stayed in the spindle for hours while the machine was not in operation.	Make sure a tool is not in the spindle when the machine is not in operation.
	The volume of air supplied to the machine is not sufficient.	Refer to: New Machine / Pre-Installation information for the correct air requirements.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct.
	The toolholder is fretting. Damage tool shank or spindle taper.	Verify the length of the tool with the manufacturer's specifications. Maintain and inspect tool holders, and spindle taper.
	There is no grease on the pull stud.	Lubricate the pull stud.
	(Through-spindle coolant only) Some coolants cause tools to stick.	Coolants containing paraffin waxes can causing sticking tools when using high pressure Through Spindle Coolant with small orifice tooling. Test with a different type of coolant.
	The spindle is getting too hot.	Test the spindle lubrication system. Lack of, or excessive oil will create heat in the spindle.
	Long gage length tooling.	Stub up the tools to the extent possible. Reduce depth of cut. Use dynamic tool path and constant cutter engagement.
While running there is bearing Noise: squealing, screeching or skating.	Noise from imbalance and vibration	Make sure the tools are balanced.
	Squealing noises from the TSC union.	Check the TSC union.
Rubbing noises; Heat at the front cap.	Spindle cartridge has been bumped there is a chance the front cap has been pushed to one side and is rubbing on the nose of the spindle shaft.	Check for clearance all the way around with a piece of 0.001" shimstock; loosen the cap, re-center and tighten.

Noises during tool change.	Some models have had the precharge circuit eliminated.	This is a reliability improvement but a noise can be heard when the TRP contacts the drawbar or the release ring. This is normal.
The spindle oscillates during spindle orientation.	The Wye contactor (if equipped) coil is not receiving power.	Make sure the Wye/Delta contactors are operating correctly. Refer to: Wye-Delta Contactor - Troubleshooting Guide .
	The spindle motor wiring is incorrect.	The spindle motor wiring is incorrect. Refer to: Spindle Motor - Wiring Configurations .
	The vector drive is defective.	The vector drive is defective. Refer to: Vector Drive - Troubleshooting Guide .
Alarm 174 Tool Load Exceeded	Tool Load limit incorrect	View the Advanced Tool Management (ATM) tab to adjust user defined tool load limit
	Aggressive feedrates	Adjust your program for a more conservative feedrate
Machine is experiencing surface finish issues	Preload set to the bearings may be incorrect	Perform the Spindle Deflection Test - Mill procedure for preload verification NOTE: Only perform this procedure <u>AFTER</u> all other troubleshooting steps outlined in the Mill Spindle Inspection Report, linked at the top of this page, have been performed.
Lube line slips out of fitting easily.	Plastic fitting has failed.	Replace fittings with updated steel version. Order PN 58-2196 .